SMART PPE DETECTION SYSTEM USING YOLOv8

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Course: Computer Vision

Project Type: Bring Your Own Project (BYOP)

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1. Introduction

I completed a Bring Your Own Project (BYOP) for my Computer Vision course, which required me to select a real-world issue and attempt to fix it using the principles I had learnt. I decided to focus on construction site safety because I wanted to concentrate on something useful and significant. Because safety concerns are so prevalent in real life, particularly on construction sites, I personally found this topic to be interesting.
If appropriate safeguards are not implemented, accidents can easily occur on construction sites. Although it is anticipated that workers wear protective gear, like as vests and helmets, this is not always done. The goal of this project is to use computer vision to determine whether employees are wearing the appropriate protective gear.

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2. Problem Statement

Safety regulations are crucial at building sites, but they are not always adequately observed. The danger of accidents is increased when workers fail to wear safety vests or helmets.
The project's goal is to develop a straightforward system that uses computer vision techniques to automatically determine whether employees are wearing personal protective equipment (PPE).

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3. Objectives

This project's primary goals are:
• To identify individuals in pictures
• To recognize protective gear such vests and helmets
• To verify adherence to safety regulations
• To comprehend how object detection functions in practical situations

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4. Methodology

I utilized a pre-trained YOLOv8 model for object recognition in this project. I concentrated more on comprehending the model's operation and using it in real-world scenarios rather than on intricate model training.

The procedure used for this project is:
• Open the YOLOv8 model.
• Give an input picture
• Recognize items such as people, vests, and helmets
• Use bounding boxes to display the results.
This technique makes it easier to see whether employees are adhering to safety regulations.

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5. Tools and Technologies Used

The technology and tools listed below were employed-

- Python OpenCV
- Ultralytics Yolov8
- NumPy

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The project's implementation and comprehension of object identification were made simpler by these tools.

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6. Implementation

In this project, I created a Python script to run the PPE detection system. The YOLOv8 model is loaded using the Ultralytics library, and an input image is passed to it.

The model processes the image and detects objects present in it. I tested the system on sample images and observed the detection results. The results are then displayed using OpenCV, where bounding boxes are drawn around detected objects. This helps in clearly visualizing whether a person is wearing safety equipment or not.

main.py > ...

```python
from ultralytics import YOLO
import cv2

# load model
model = YOLO("models/yolov8n.pt")

# load image
img = cv2.imread("source_files/construction-safety.jpg")

# prediction
results = model(img)

# show result
for r in results:
    frame = r.plot()

cv2.imshow("PPE Detection", frame)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Figure 2: Code snippet showing model loading and detection

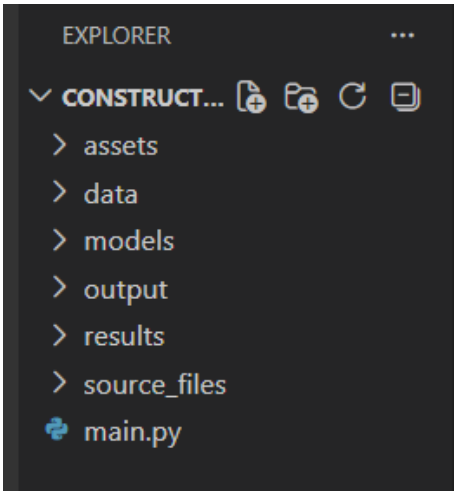

Figure 3: Project folder structure

7. Results

The image's items were correctly identified by the system, which then displayed them with bounding boxes. It recognized individuals and protective gear such as vests and helmets.
It is simple to determine whether safety precautions are being taken thanks to the output.

Figure 1: Output showing detection of person and PPE

8. Challenges Faced

While working on this project, I faced a few challenges:

- Setting up the required libraries
- Understanding how to run the YOLO model
- Fixing errors during execution
- Managing file paths correctly

I was able to solve these issues step by step and complete the project successfully. This project gave me confidence that even with basic knowledge, I can build useful applications using AI.

9. Key Learnings

This project helped me learn:

- How object detection works in real-world applications
- How to use pre-trained models like YOLOv8
- How computer vision can be applied to safety problems
- Basic usage of OpenCV for displaying results

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10. Future Scope

This project can be improved further in the following ways:

- Extending it to real-time video detection
- Improving accuracy with better training
- Adding detection for more safety equipment
- Deploying it as a real-time monitoring system

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11. Conclusion

This project demonstrates how computer vision can be used to solve real-world problems like construction site safety. Even a simple system like this can help improve awareness and safety compliance.

Working on this project helped me gain practical knowledge and confidence in applying AI concepts.

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12. References

- Ultralytics YOLOv8 Documentation
- OpenCV Documentation
- Online resources related to PPE detection